



BLITTA – TOGO SOLAR POWER PLANT



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Project Title	<i>BLITTA – TOGO SOLAR POWER PLANT</i>
Version	<i>02.1</i>
Report ID	<i>BELL_VCS_VAL_VER_11922</i>

Report Title	<i>BLITTA – TOGO SOLAR POWER PLANT</i>
Client	<i>ClimatePartner GmbH</i>
Pages	<i>36</i>
Date of Issue	<i>15-August-2023</i>
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Summary:

Verification purpose: The current project is 70 MW solar power project in Blitta, Togo promoted by AMEA Solar Togo SAU. The project is registered with VERRA registry with VERRA ID - 3298¹. The project activity generates renewable energy which is exported to the national grid of Togo. The project has contributed to reduction of GHG emissions by displacing fossil fuel dominated grid-based electricity with renewable energy – solar energy based renewable electricity.

Verification is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of verified carbon units (VCUs).

The scope of the verification is the independent and objective review of the monitoring report (MR). The MR is reviewed against the relevant criteria (see above) and decisions by the CDM Executive Board and VCS executive board, including the approved baseline and monitoring methodology. The verification was based on the guidance given in the CDM validation and verification standard for project activities version 03.0, reviewed against registered VCS-PD and final validation report, VCS program guide version 4.3 and VCS standard version 4.4.

The assessment team has employed a risk-based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the MR. The main focus of the assessment team is to identify the significant risks for the project implementation and the generation of VCUs. The verification is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the monitoring report combined.

The start date of the project activity is 12-June-2021. An undertaking has been submitted by PP for double counting confirming that no GHG reduction will be claimed in any other GHG mechanism for current monitoring period. The project activity has observed continued operation and no major breakdown has taken place except routine maintenance/shutdown.

This is 1st verification and certification for the greenhouse gas (GHG) emission reductions reported from 'BLITTA – TOGO SOLAR POWER PLANT' for the monitoring period 12-June-2021 to 31-December-2022 (Inclusive of both days), under the crediting period 12-June-2021 to 11-June-2028. Total emission reductions achieved for the

¹ <https://registry.verra.org/app/projectDetail/VCS/3298>

project activity during the current monitoring period are 70,969 tCO₂e. An undertaking from the project participant confirms that project will not claim any other scheme benefits for the concerned monitoring period.

A risk-based approach has been followed to perform this verification activity. In the course of verification, 05 Corrective Action requests (CAR), 01 Clarification Request (CL) and 00 Forward action requests (FARs) were raised and successfully closed. The review of the monitoring report and additional documents related to baseline and monitoring methodology; the subsequent background investigation, on-site visit, follow-up interviews with project promoter and project participants have provided VVB with sufficient evidence to verify the fulfilment of the stated criteria of VCS. Since an on-site visit was conducted and all the documents were checked, VVB ensured that no uncertainties are involved with this verification.

LGAI Technological Center S.A. (Applus+ Certification) (Hereafter referred to as Applus+ Certification) has been appointed by 'ClimatePartner GmbH²' to perform 1st verification of the "BLITTA – TOGO SOLAR POWER PLANT" under VCS standard, version 4.4 and VCS program guide, version 4.3. The objective of this verification activity is to have an independent third party for the assessment of the project design, monitoring report and Final Verification report and to ensure a thorough assessment of the project activity against the applicable CDM and VCS requirements. In particular; the project's baseline, monitoring plan is assessed against the applied methodology "ACM0002 version 20.0 – Grid-connected electricity generation from renewable resources. The project's compliance with the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS program guide 4.3, VCS standard, version 4.4 and CDM validation and verification standard for project activities, version 3.0.

The only purpose of the verification is its usage during the issuance process as part of the VCS project cycle. Therefore, LGAI Technological Center S.A. (Applus+ Certification) can't be held liable by any party for decisions made or not made based on the verification opinion, which will go beyond that purpose.

The verification has been planned and organized to achieve a reasonable level of assurance as per the requirement of VCS. No sampling procedure applied for site visit or document verifications. The entire documents checked/on-site plant verification conducted to arrive at positive verification conclusions.

² A contract is signed was ClimatePartner GmbH who is a project participant in a role of consultant for the project.

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1 INTRODUCTION

1.1 Objective

LGAI Technological Center S.A. (Hereinafter referred as Applus+ Certification) has been appointed by ClimatePartner GmbH to perform 1st verification of the project “BLITTA – TOGO SOLAR POWER PLANT” under VCS standard version 4.4 and VCS program guide version 4.3. The objective of this verification activity is to have an independent third party for the assessment of the project design, monitoring report and final verification report and to ensure a thorough assessment of the project activity against the applicable CDM and VCS requirements. In particular;

- the project’s baseline and monitoring plan is assessed against applied methodology – ACM0002 (version 20.0) “Grid-connected electricity generation from renewable sources”
- VCS program guide, version 4.3 and VCS standard, version 4.4
- The project’s compliance with the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria
- CDM validation and verification standard for project activities, version 03.0

Verification is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified carbon units (VCUs).

1.2 Scope and Criteria

The scope is defined as, an independent and objective review of the Monitoring report (MR) prepared as per the registered VCS PD and registered approved methodology ACM0002 version 20.0. The MR is reviewed against the criteria stated in Article 12 of the Kyoto Protocol, the CDM modalities and procedures as agreed in the Marrakech Accords and the relevant decisions by the CDM Executive Board and VCS guidelines, including the approved baseline and monitoring methodology applied. The verification was based on the requirements in the CDM validation and verification standard for project activities, version 03.0, VCS program guide, version 4.3 and VCS standard, version 4.4.

The verification is not meant to provide any consulting towards the project participant. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the Monitoring report. In line with Guidelines for Application of materiality in verifications, the verification team has conducted a complete verification of all the information presented in the monitoring report and data monitored as presented in the emission reduction calculation spread sheet. There are no material errors, overestimation of emission reduction, omission or misstatement. The verification team has reviewed the documents like commissioning certificate, daily generation reports, monthly invoices etc.

1.3 .Level of Assurance

The verification has been planned and organized to achieve a reasonable level of assurance as per the requirement of VCS and PP's selection. The entire set of documents have been checked/on-site verification conducted to arrive at positive verification conclusion.

The verification strategy used by Applus+ Certification is based on an understanding of the risks involved in reporting GHG emission data and the mitigation measures put in place. verification team has verified the monitoring data for all the parameters of the monitoring plan and confirms that the reported emission reductions are free from any type of material errors or misstatements.

1.4 Summary Description of the Project

The Blitta - Togo solar PV plant is one of the largest in West Africa and the first renewable energy facility in the country pf Togo. The plant commissioned by Amea Solar Togo SAU supplies reliable, clean electricity to the national grid. The current capacity of the plant is 50 MW. An additional 20 MW expansion is expected to be operational by 2024 bringing the total to 70 MW. VVB confirmed that the expansion is ongoing during the on-site visit.

During the Current Monitoring Period from 12-June-2021 to 31-December-2022 (First and last date included) the project activity has supplied 123,855.67 MWh of electricity contributing to 70,969 tCO_{2e} of GHG reductions.

2 VERIFICATION PROCESS

2.1 Method and Criteria

Verification Process

The project assessment is based on the Clean Development Mechanism validation and verification standard for project activities, version 03.0 and VCS standard 4.4 and VCS program guide version 4.3 and is conducted using standard auditing techniques to assess the correctness

of the information provided by the project participants. Before the assessment begins, members of the team covering the technical scope(s), sectoral scope(s), and relevant host country experience for evaluating the VCS project activity are appointed.

Once the project is received by the assessment team, the members of the assessment team carried out:

- A desk review of the monitoring Report against the registered PD;
- On-site visit and follow-up interviews with project participant and stakeholders;
- The resolution of outstanding issues and the issuance of the final verification report and opinion.

The prepared verification report and other supporting documents then undergo an internal quality control at the HQ (Accredited office) before being submitted to the VCS executive board.

In order to ensure transparency, assumptions must be clear and stated explicitly and background material must also be referenced. LGAI Technological Center, S.A. (Applus+ Certification) has developed a specific checklist customized for the project. The checklist demonstrates, in a transparent manner, the project criteria (requirements), discussion on each criterion by the assessment team, and the results from verifying the identified criteria.

Appointment of the assessment team

According to the sectoral scope / technical area and experience in the sectoral or national business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification).

The composition of audit team shall be approved by the LGAI Technological Center S.A. (Applus+ Certification) ensuring that the required skills are covered by the team.

The four qualification levels for team members that are assigned by formal appointment rules are as presented below:

Lead Auditor (LA).

Observer (Obs)

Technical Expert (TE)

Technical Reviewer (TR)

The sectoral scope / technical area knowledge linked to the applied methodology/ies shall be covered by the assessment team.

Name	Role	SS Coverage	TA Coverage	Financial aspect	Host country experience
Deepak Pundlik	LA/TE	YES	YES	NA	Yes
Srikanth Meesa	TR	YES	YES	NA	NA

The complete list of CVs is included as Appendix 3 of this report.

Document review

The Monitoring Report submitted by the project developer was reviewed against the approved methodology, registered VCS PD, final validation report and other relevant criteria to verify the correctness, credibility, and interpretation of the presented information. Furthermore, a cross-check between information provided and information from other sources has been done. A complete list of all documents and evidence material reviewed is included in this report below in appendix 1.

On-site visit

An on-site visit was conducted by LGAI Technological Center S.A. (Applus+ Certification) who performed interviews, project activity verification with project stakeholders to confirm selected information and to resolve issues identified in the document review. The details are provided in the below sections.

Resolution of Clarification and Corrective Action Request

The objective of this phase of the verification was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for Applus+ Certification positive conclusion on the monitoring report. The Corrective Action Requests and Clarification Requests raised by Applus+ Certification were resolved during communications between the Client and Applus+ Certification to guarantee the transparency of the verification process, the concerns raised and responses given are summarized below in the Appendix 2.

The final MR submitted by PP serves as the basis for the final assessment presented. Additional changes to the project during the verification process are not considered to be significant with respect to the main VCS objectives i. e. reduction of anthropogenic GHG emissions and contribution of sustainable development to the host country.

Internal quality control

As final step of a verification of the final documentation including the final verification report and the checklist have to undergo an internal quality control by the technical review committee, i.e. each report has to be finally approved either by the head of the technical review committee or the deputy. In case one of these two persons is part of the assessment team approval can only be given by the other one to avoid any conflict of Interest.

After confirmation of the project developer, the positive verification opinion and relevant documents are submitted to the VCS secretariat through the VCS web-platform.

VVB has provided below the verification schedule, including key milestones:

Desk review of documents: 05-June-2023 to 08-June-2023

On-site visit: 09-June-2023

Draft Verification Report: 10-June-2023

2.2 Document Review

The Monitoring Report submitted by PP was reviewed against the approved methodology, registered VCS PD, final validation report and other relevant documents and criteria to verify the correctness, credibility, and interpretation of the presented information. Furthermore, a cross-check between information provided and information from other sources has been done. A complete list of all documents and evidence material reviewed is included in this report below in appendix 1.

2.3 Interviews

An on-site audit was conducted for the project activity on 09-June-2023.

During the visit, VVB verified technical details & metering/monitoring arrangement, energy generation reports, calibration certificates etc. All the documents were cross checked to ensure conservative estimation of emission reductions.

The PP representatives were questioned about the implementation of the project activity. Several topics like the verification of commissioning date, generation records, monitoring of the data and the error accountability were discussed. To cross check the information provided by PP, various documents like technical specifications, commissioning certificates, power purchase agreements, energy generation records, invoices, calibration certificates were also verified. The names of the persons interviewed is given below;

Organization	Name of Persons/Designation	Topics discussed	Team Member
Herve Rloux	Amea Solar Togo SAU – Site incharge	Project activity implementation	Deepak Pundlik
Rauny Damtare	Amea Solar Togo SAU – EHS manager	Operation, Calibration, O&M practices, energy generation records etc. LSC mechanism, Mechanical maintenance, Electrical maintenance, Project description, Invoicing practices	
K Manikandan	Jakson Power – Site incharge		
Mousa Djobiut	Local stakeholder (BLITTA)	Project operation, its impact on the local area, project's impact on sustainable development, any grievances	
Jannane Jameteve	Local stakeholder (BLITTA)		
Amer Arafat	Element Six - Partner	MR, ER calculations etc.	

Local stakeholders were questioned about the project operation, its impact on the local area, project's impact on sustainable development in the region and their grievances, if any. No grievances were recorded by the interviewed stakeholders.

2.4 Site Visits

The site visit was conducted by TL of the project on 09-June-2023 at the project site in Blitta, Togo. During the site visit, VVB performed following activities.

1. Assessment team checked the implementation of the project, Baseline emission, Emission reduction calculation, technical description of the project and Monitoring.
2. Assessment team also checked that whether the monitoring plan as described in the VCS PD is actually practised onsite.
3. Assessment team also checked any change in host country criteria which may affect the baseline of the project activity.
4. Assessment team also checked procedure for on-going stakeholder communication and means of communication available for the same i. e. grievance register present at the site

2.5 Resolution of Findings

The objective of this phase of the verification was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification)'s positive conclusion on the project design and Monitoring report. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communication between the client and LGAI Technological Center S.A. (Applus+ Certification) to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in the appendix 2.

The final MR submitted serves as the basis for the final assessment presented. Additional changes to the project during the verification process are not considered to be significant with respect to the main VCS objectives. The two main VCS objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of verification findings	No. of CL	No. of CAR	No. of FAR
Project Document and Monitoring report	--	02	--
Description of project activity	01	---	--

Areas of verification findings	No. of CL	No. of CAR	No. of FAR
Application of selected baseline and monitoring methodology and selected standardized baseline			
Applicability of methodology and standardized baseline	--	--	--
Deviation from methodology	--	--	--
Clarification on applicability of methodology, tool and/or standardized baseline	--	--	--
Project boundary	--	--	--
Establishment and description of baseline scenario	--	--	--
Demonstration of additionality	--	--	--
Emission reductions	--	--	--
Calibration details	--	01	--
Monitoring plan	--	01	--
No Net harm assessment	--	--	--
Local stakeholder consultation	--	--	--
Others (please specify)- - ER Sheet	--	01	--
Total	01	05	00

The list of findings and their resolution is presented in appendix 2 of this report.

2.5.1 Forward Action Requests

This is 1st verification of the project activity and no FAR was raised from validation and during this verification process.

2.6 Eligibility for Validation Activities

This section is not applicable for present verification, as Applus+ Certification holds the accreditation for Validation of projects under this Sectoral Scope.

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

The project activity was registered under VERRA with project ID – 3298 and with 1st crediting period from 12-June-2021 to 11-June-2028.

As per the MR and undertaking provided, PP would not consider the credit from any other mechanism for the current monitoring period. The undertaking is provided to confirm that there is no any double accounting for current monitoring period. Further, assessment team confirms that the project activity is not participating in any other form of environmental credits.

3.2 Methodology Deviations

The project activity has applied correct methodology which are as per the registered VCS PD and thus no deviation is sought regarding the methodology. The project complies with all the requirement of the methodology and thus deviation to the methodology is not a requirement for the present project activity.

3.3 Project Description Deviations

The project does not involve any project description deviation.

3.4 Grouped Project

The project does not involve any addition of new project activity and thus the project do not fall under grouped project.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

During the on-site visit, it was confirmed that the project is implemented as per registered VCS PD which is verified from commissioning certificate, statutory clearances submitted, energy generation records. During the current monitoring period it was observed that no unforeseen situation evolved which can impact the operation of the project activity. Scheduled maintenance was carried out as per the instruction of the manufacturer and the same is acceptable to the assessment team.

Project location is confirmed by the assessment team through GPS co-ordinates using google earth software and are same as mentioned in the registered VCS PD and are correct. The geographical co-ordinates are 8° 21'52.33"N, 1° 1'4.42"E and the project site is located near Blitta town, Togo.

The project was commissioned on 12-June-2021 which is the Project Start date as confirmed based on the commissioning certificate submitted and is in line with VCS standard version 4.4. Total project capacity is 70 MW and remaining 20 MW plant is still under construction as confirmed during the site visit.

The technical specification of the project activity is as follows which is confirmed during site visit based on technical datasheet:

Solar Modules:

Module Type: JKM390M-72-V

Maximum Power (Pmax): 390/395/400 Wp

Module technology: Poly Crystalline

No. of modules: 127,482

Inverters:

Model: Sungrow SG250HX- 250KTL – H1 String or equivalent

No. of Inverters: 200

Technical Lifetime of the plant: 25 years

VVB confirms that there are no material misstatements between the actual monitoring system, and the monitoring plan set out in the project description and the applied methodology.

The assessment team confirmed through interview with O & M personnel (Jakson Power) that there are no changes into the project design during this monitoring period. It was found that the monitoring plan was implemented as per the requirement of the VCS PD, approved monitoring plan and applied methodology. The organisational role and responsibility as mentioned in the registered VCS PD is followed onsite. The energy meters used are calibrated periodically and calibration is not due. All the emergency preparedness as mentioned in the registered VCS PD is followed onsite and no discrepancies were found regarding the same. PP has maintained stakeholder grievance register at the site which was checked to confirm that no negative comments with respect to the Project operation were received during current monitoring period. During the current monitoring period, there were no major shutdown/break down observed for the project activity as confirmed based on plan register/interview with O & M personnel during the on-site visit and the VVB noted that the plant was in continual operation.

The Project developer contribution from the project activity towards sustainable development is in accordance to host country.

The project displaces electricity produced in fossil fuel-based power plants, by using clean renewable source to generate electricity and thereby contributes to sustainable development through reduction in greenhouse gas (GHG) emissions. It has also been confirmed that the project implementation has led in contribution to SDG 03, 04, 07, 08 and SDG 13 (as stated in section 1.11 of the submitted MR). The project supplied clean, affordable and renewable energy (123,855.67 MWh net electricity under the monitoring period) without reliance on fossil fuels for generation of electricity to meet the growing demands in the region. The generated electricity avoided release of 70,969 tCO₂ into the atmosphere.

The project is not involved in any other form of GHG emission program and VCU's generated from this verification will not be used for other trading program to avoid any kind of double counting. The same is confirmed by the PP representative during the verification audit. Assessment team also conducted independent review regarding the same and found that the statement of the PP is accurate and project is not involved in any other kind of GHG trading for the present monitoring period and further the project has received or sought any other form of environmental credit, or has become eligible to do so since validation or previous verification.

The assessment team observed that the project is in line with the registered VCS PD and applied methodology. PP has not applied for project deviation in current monitoring period.

The current project is not in a supply chain and hence no public statements needed by the producer(s), retailer(s) or project proponent (as applicable) saying that VCUs may be issued for the GHG emission reductions and removals associated with the impacted goods and services.

The project has not applied for any previously validated methodology deviations.

Assessment team confirms following during the verification audit:

1. Start date of the project is 12-June-2021.
2. An undertaking letter has been submitted by PP for no double counting with any other GHG program. PP also has given a written declaration that project will not claim other form of GHG credit for the concerned monitoring period.

3. Assessment team confirms that this is 1st monitoring under VCS and covers the activity from 12-June-2021 to 31-December-2022 (inclusive of both dates). VCS crediting period is of 10 years with 12-June-2021 as the start date and crediting period end date as 11-June-2028 (both days included).
4. GHG credits from 12-June-2021 to 31-December-2022 are claimed under VCS only. At any point of time during the current crediting period, the project proponent has abide by the "Double Counting".
5. Assessment team checked and found that the Project proponent of the project activity are as below for the current monitoring period:

Organization name	AMEA Solar Togo SAU
Contact person	Basma Bentaher
Title	Senior Director – Project Development
Address	Marina Plaza Offices 3301/3302/3303 Level 33 Dubai Marina PO Box 37669 Dubai United Arab Emirates
Telephone	+971 4 310 7014
Email	Basma.Bentaher@ameapower.com

Organization name	ClimatePartner GmbH
Contact person	Dr. Sascha Lafeld
Title	Head of Carbon Offsets and Green Energy Services
Address	St.-Martin-Str. 59, D-81669 Munich, Germany
Telephone	+49 89 1222 8750
Email	sascha.lafeld@climatepartner.com

The other entities involved in the project activity are as below:

Organization name	Element 6 – FZCO
Role in the Project	Consultant
Contact person	Amer Arafat
Title	Partner
Address	Technohub 1, DTEC, Dubai Silicon Oasis, Dubai, UAE
Telephone	+971 52 938 9590
Email	amer@element6.cc

The quantified emission reduction calculation for the monitoring period is correct and conservative. Assessment team also compared actual VCUs with the estimated VCUs. The project is commissioned on 12-June-2021. Considering actual working days, the estimated emission reductions are 110,807 tCO₂e. The actual emission reductions are 70,969 tCO₂e which are 35.95% lower than the estimated emission reductions and are acceptable as the plant is not yet fully commissioned with its total installed capacity i. e. 70 MW.

Finding: CL 01 and CAR 01, 02 were raised and successfully closed. Please refer Appendix 2 for more information.

4.2 Safeguards

4.2.1 No Net Harm

No potential environment or socio-economic matter was found during the discussion with PP representatives during the audit. The project is renewable energy project and thus no negative impact is envisaged on-site and as confirmed by PP.

The strategic environmental and social assessment report was checked which clearly mentioned that solar PV power plant does not result in direct air pollution, noise and water pollution.

There was no harm identified from the project activity and hence no need to apply mitigation measures. Hence verification team confirms that there are no significant impacts on air, water, soil quality and ambience due to the project activity.

The project activity has no potential negative environment and social-economic impacts which is confirmed from the remote interview of the local people (refer section 2.3 of this report). The project activity creates positive socioeconomic impacts by providing jobs to local people as confirmed from the remote interview of the local people and from the supportive documents.

4.2.2 Local Stakeholder Consultation

During on-site visit, VVB checked and confirmed that a grievance register is placed at the office of PP on the gate of the project activity. As confirmed from the external local stakeholders, no grievances were raised for the current monitoring period. The internal stakeholders i. e. workers at the project site had raised some grievance which was addressed by PP through their O & M firm and successfully closed. In appendix II of MR, PP has provided the concerned grievance log and its closure.

PP has mentioned grievances received from internal/external stakeholders in appendix II of VCS MR which was checked during the on-site visit and confirmed. Thus, VVB noted that ongoing communication with local stakeholders is possible and is being taken care by PP.

4.3 AFOLU-Specific Safeguards

The current project activity is non-AFOLU projects, hence this section is not required.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

Means of verification	The verification team assessed whether the data and calculations of GHG emission reductions achieved resulting from the MR are accurate. The verification team has checked whether calculations of baseline GHG emissions, project GHG emissions and leakage GHG emissions have been carried out in accordance with the formulae and methods described in the monitoring plan of the VCS PD and MR.								
Findings	CAR 03 and 04 were raised and successfully closed. Please refer Appendix 2 for more information.								
Conclusion	Baseline emission: The baseline Emissions for a given year is calculated by multiplying the energy generation (EG) with the grid emission factor of the grid. Formula Used: - $BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$ Where; BE_y= Baseline emission in tCO₂e $EG_{PJ,y}$ = Quantity of net electricity generation supplied to the grid during the monitoring period by the solar plant (MWh/yr) $EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using “TOOL07: Tool to calculate the emission factor for an electricity system”, version 07 (tCO₂/MWh) Ex-ante parameters: <table><tr><th>Parameter</th><th>Description</th><th>Source/Justification</th></tr><tr><td>$EF_{grid,CM,y}$ tCO₂/MWh</td><td>Combined Margin CO₂ grid emission factor in the year y</td><td> Calculated as per “Tool to calculate the emission factor for an electricity system, version 07” PP has considered the value as per Standardized Baseline ASB0034-2021 Version 01.03 which is in line with the approved VCS PD. The value 0.573 is used for the calculation of the Baseline Emissions and is found to be consistent with the registered VCS PD. </td></tr></table>			Parameter	Description	Source/Justification	$EF_{grid,CM,y}$ tCO ₂ /MWh	Combined Margin CO ₂ grid emission factor in the year y	Calculated as per “Tool to calculate the emission factor for an electricity system, version 07” PP has considered the value as per Standardized Baseline ASB0034-2021 Version 01.03 which is in line with the approved VCS PD. The value 0.573 is used for the calculation of the Baseline Emissions and is found to be consistent with the registered VCS PD.
Parameter	Description	Source/Justification							
$EF_{grid,CM,y}$ tCO ₂ /MWh	Combined Margin CO ₂ grid emission factor in the year y	Calculated as per “Tool to calculate the emission factor for an electricity system, version 07” PP has considered the value as per Standardized Baseline ASB0034-2021 Version 01.03 which is in line with the approved VCS PD. The value 0.573 is used for the calculation of the Baseline Emissions and is found to be consistent with the registered VCS PD.							

³ https://cdm.unfccc.int/methodologies/standard_base/2015/sb166.html

	Monitored Parameter:													
	<table><tr><th>Parameter</th><th>Description</th><th>Source/Justification</th></tr><tr><td>EG_{PJ,y}</td><td>Quantity of net electricity generation supplied to the grid during the monitoring period by the solar plant</td><td> Net electricity exported to the grid is calculated based direct measurement of electricity according to daily generation reports and invoices raised by PP. The calculation presented in the ER sheet confirms the net electricity export to the grid. PP has considered both the data i. e. daily generation report as well as invoices raised by PP. PP has considered minimum of the value between energy generation reports and invoices for emission reduction calculation and same is accepted. Due to delay in calibration, PP has applied error factor for the delayed period in line with applicable guidance. The monitored value for the current period is 123,855.67 MWh. </td></tr></table>					Parameter	Description	Source/Justification	EG _{PJ,y}	Quantity of net electricity generation supplied to the grid during the monitoring period by the solar plant	Net electricity exported to the grid is calculated based direct measurement of electricity according to daily generation reports and invoices raised by PP. The calculation presented in the ER sheet confirms the net electricity export to the grid. PP has considered both the data i. e. daily generation report as well as invoices raised by PP. PP has considered minimum of the value between energy generation reports and invoices for emission reduction calculation and same is accepted. Due to delay in calibration, PP has applied error factor for the delayed period in line with applicable guidance. The monitored value for the current period is 123,855.67 MWh.			
	Parameter	Description	Source/Justification											
	EG _{PJ,y}	Quantity of net electricity generation supplied to the grid during the monitoring period by the solar plant	Net electricity exported to the grid is calculated based direct measurement of electricity according to daily generation reports and invoices raised by PP. The calculation presented in the ER sheet confirms the net electricity export to the grid. PP has considered both the data i. e. daily generation report as well as invoices raised by PP. PP has considered minimum of the value between energy generation reports and invoices for emission reduction calculation and same is accepted. Due to delay in calibration, PP has applied error factor for the delayed period in line with applicable guidance. The monitored value for the current period is 123,855.67 MWh.											
	The verification team has checked the entire energy generation records applicable for the monitoring period and found the monitoring parameter is monitored and recorded as per the monitoring plan in the registered VCS PD.													
	The verification team has crosschecked the emission reduction sheet and monitoring report data with the energy generation reports and found all the values are matching. These values were further cross checked with the invoices submitted.													
The calculation of baseline emissions is as below,														
BE_y= EG_{PJ,,y} x EF_{grid,CM,y} BE_y = 123,855.67 x 0.573 BE_y = 70,969 tCO_{2e} (round down values)														
<table><tr><th>Year</th><th>Baseline emissions or removals (tCO_{2e})</th><th>Project emissions or</th><th>Leakage emissions (tCO_{2e})</th><th>Net GHG emission reductions or</th></tr><tr><td></td><td></td><td></td><td></td><td></td></tr></table>					Year	Baseline emissions or removals (tCO _{2e})	Project emissions or	Leakage emissions (tCO _{2e})	Net GHG emission reductions or					
Year	Baseline emissions or removals (tCO _{2e})	Project emissions or	Leakage emissions (tCO _{2e})	Net GHG emission reductions or										

			removals (tCO ₂ e)		removals (tCO ₂ e)
	12-June-2021 to 31- December- 2021	24,402	0	0	24,402
	01-January- 2022 to 31- December- 2022	46,567	0	0	46,567
	Total	70,969	0	0	70,969
Project emissions: The project activity involves harnessing solar power. Hence, in line with the applied methodology, project emissions are zero. Leakage emissions: The project activity involves harnessing solar power. Hence, in line with the applied methodology, leakage emissions are zero. Thus, Emission Reductions are: The emission reductions (ER_y) by the Project activity during a monitoring period is the difference between baseline emissions (BE_y), project activity emissions (PE_y), as follows $ \begin{aligned} ER_y &= BE_y - PE_y \\ &= 70,969 - 0 \\ &= 70,969 \text{ tCO}_2\text{e} \end{aligned} $ VVB confirmed that GHG reductions and removals have been quantified correctly and in line with monitoring procedure in VCS PD and applied methodology.					

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

Means of verification	The verification team checked the shutdown/break down log for the monitoring period. During the verification audit, the energy meters were also checked.
Findings	No findings raised.

Conclusion	The Project activity is operated and managed by Amea Solar Togo SAU and Jackson Power is the O&M contractor for the solar project. As confirmed during the audit, the net energy export is recorded in the generation reports online and daily generation reports are generated which were made available to VVB and checked during the audit. The current energy meters were installed at the time of project commissioning and were pre-calibrated on 10-October -2020, details about which were checked based on calibration certificates submitted and details are presented in appendix 5. The energy meters of accuracy class of 0.2s. Based on the submitted documents, VVB confirmed that the value of net electricity exported to the grid as used in emission reduction calculation is correct. Delayed calibration is observed for all the energy meters during the year 2021 and PP has applied error factor for the delayed period as the delayed calibration confirmed that identified error is below the maximum permissible error. It is reported that the data will be kept for 2 years following the end of the crediting period or till the last issuance of VCUs for the project activity whichever occurs later. The responsibilities and authorities of project management, data handling and recording, measurement methods and QA/QC procedure have been systematically established and formalized and the same was verified during discussion with PP representative. Interview with site personnel also confirmed that the operational and organizational chart as mentioned in registered VCS PD is as per the site practice and thus assessment team confirms that the details are correct. PP has submitted shutdown / break down reports and found that the plant undergone scheduled maintenance. VVB checked the routine maintenance log book and confirmed that it does not have any impact on project design and monitoring procedures. No unforced error observed. PP has established internal auditing procedure as mentioned in the MR which was checked and confirmed during the on-site visit. Assessment team confirmed that data/ information used for determining GHG reductions and removals were sufficient in quantity and of appropriate quality. QA/QC procedure was checked and found to be appropriate.
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4.6 Non-Permanence Risk Analysis

No.	Risk that could lead to material errors, omissions or misstatements	Assessment of the risk		Response to the risk in the verification plan and/or sampling plan
		Risk level	Justification	
1.	Not applicable	Not applicable	Not applicable	Not applicable

5 VERIFICATION OPINION

The GHG statement is the responsibility of the project proponent, whether the project conforms with the verification criteria for projects and their GHG emission reductions or removals set out in VCS Version 4, including any qualifications or modifications, and a description of the reason for qualifications or modifications placed before the conclusion.

VVB confirms that the project has been implemented in accordance with the project description.

Applus+ Certification has been engaged by ClimatePartner GmbH to perform 1st verification of the “BLITTA – TOGO SOLAR POWER PLANT”.

The VCS Standard, version 4.4, VCS Program Guide, version 4.3, and Registration & Issuance Process (version 4.3) were used as the foundation for the verification process.

Our Verification approach was based on the requirements as defined under the Kyoto Protocol, Marrakesh accord, as well as those defined by the CDM Executive Board and VCS program guide and VCS standard. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification team can confirm that:

- the project is operated as planned and described in the project document;
- the monitoring plan is as per the applied methodology;
- the monitoring process in monitoring report is as per the VCS PD
- the development and maintenance of records and reporting procedures are in accordance with the monitoring plan
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.
- A reasonable level of assurance was achieved during the verification.
- No limitation observed for the present verification

Verification Analysis: The verification strategy used by Applus+ Certification is based on an understanding of the risks involved in reporting GHG emission data and the mitigation measures put in place. In order to provide a reasonable level of assurance that reported GHG emission reductions are accurately stated, Applus+ Certification prepared and carried out the verification by gathering documentation, additional information, and justifications.

The GHG statement is the responsibility of the project proponent and the project conforms with the validation and verification criteria for projects and their GHG emission reductions or removals set out in VCS Version 4.

VVB declares that the GHG statement verification was conducted in accordance with ISO 14064-3: 2019.

Verification period: *From 12-June-2021 to 31-December-2022 (first and last date included).*

Verified GHG emission reductions and removals in the above verification period, broken down by calendar year:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
12-June-2021 to 31-December-2021	24,402	0	0	24,402
01-January-2022 to 31-December-2022	46,567	0	0	46,567
Total	70,969	0	0	70,969

Considering actual working days, the estimated emission reductions are 110,807 tCO₂e. The actual emission reductions are 70,969 tCO₂e which are 35.95% lower than the estimated emission reductions and are acceptable as not much difference is observed.

Year	Ex-ante emissions reductions/removals	Achieved emissions reductions/removals	Percent difference	Justification for the difference
12-June-2021 to 31-December-2021	40,823	24,402	40.23%	The project is not yet fully commissioned.
01-January-2022 to 31-December-2022	69,983	46,567	33.46%	The project is not yet fully commissioned.
Total	1,10,807	70,969	35.95 %	

APPENDIX 1: DOCUMENTS REVIEWED DURING VERIFICATION

No.	Author	Title	References to the document	Provider
1.	NA	Monitoring Report, (Initial Version) Monitoring Report, (Final Version)	Version 01, dated 04-June-2023 Version 05 dated 21-July-2023	PP
2.	NA	Emission Reduction sheet (Initial Version) Emission Reduction sheet (Final Version)	Version 01, dated 04-June-2023 Version 03 dated 21-July-2023	PP
3.	NA	Registered VCS PD Validation report	Version 07 dated 11-April-2023 Version 2.2, dated 12-April-2023	VERRA
4.	NA	VCS Standard VCS Program guide VCS Validation Verification Manual	Version 4.4 Version 4.3 version 3.2	
5.	NA	Technical specifications / photographs of solar project, electricity meters etc. Single line diagram and meter locations / Layout		VERRA
6.	NA	Power purchase agreement between PP and Togo electrical energy company	02-August-2020	PP
7.	NA	Operation and maintenance agreement between PP and Jakson Energy		PP
8.	NA	Performance test acceptance report by Togo electrical energy company	01-June-2021	PP
9.	NA	ACM0002: Grid-connected electricity generation from renewable sources CDM validation and verification standard for project activities	Version 20.0 Version 3.0	UNFCCC
10.	NA	Name plate pictures for PV modules, Inverters, transformer and meters		PP
11.	NA	Declaration by PP for No double counting of Emission Reductions	03-November-2022	PP
12.	NA	Electricity daily generation reports for electricity supplied to grid for this monitoring period and invoices raised		PP
13.	NA	Organization structure/chart		Other
14.	NA	Supporting documents for		PP

No.	Author	Title	References to the document	Provider
		creation of local jobs, local infrastructure development, trainings to local etc.		
15.	NA	Grievance register		PP
16.	NA	Energy meter calibration certificates		PP
17.	NA	On-site visit photographs		PP

APPENDIX 2: CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS(CAR/CL/FAR)

CL ID	01	Section no.	3.1	Date : 10-June-2023
Description of CL				
1. PP shall provide declaration towards no participation under other GHG program and claiming other form of credits. 2. PP shall provide electricity generation records, invoices raised against the same and energy meter calibration certificates to confirm the ER calculation.				
Project participant response				Date: 14-June-2023
1. Declaration towards no participation under other GHG program and claiming other form of credits has been provided. Please refer to the link shared in the email. 2. Electricity generation records, invoices raised against the same and energy meter calibration certificates have been provided. Please refer to the link shared in the email.				
Documentation provided by project participant				
Togo SPV Undertaking v 2_Carbon Credits Registration_2022 11 02 clean.pdf TOGO_AMEA_Invoices.pdf Calibration Evidence Folder				
VVB assessment				Date: 22-June-2023
1. PP has submitted declaration dated 03-November-2022 which confirms that the project has not participated in other GHG program and will not claim credits from other program. VVB has checked GHG registry such as CDM, GS4GG, GCC and I-Rec to confirm that the project is not registered the same which is confirmed. Hence this part of CL is closed. 2. PP has submitted electricity generation records in form of DGR records in excel worksheet. There is discrepancy observed in the values presented in DGR record, invoice/ER sheet. E. g. for the month of June, 21, energy generation as per DGR records from 12th June, 2021 to 30th June, 2021 is 11,576.60 MWh (sum of data for June, 21 in column W of worksheet – DGR sheet_TOGO.). The energy exported from 12th June, 2021 to 30th June, 2021 as per the invoice is 3,811.73477 MWh. The ER sheet provides this value as 3,811.74 MWh. PP shall clarify the same. PP has submitted copies of calibration certificate which were checked and found appropriate. PP shall provide this calibration records in the VCS MR for easy reference of a 3rd party (VERRA) and for future verifications. Hence this part of CL is open. CL is open.				
Project participant response				Date: 07-July-2023
2. The DGR sheet (DGR sheet_TOGO) that VVB is referring to was provided in July 2022 for validation purposes only. It was not submitted as a part of the verification and hence was not the updated				

version. The previous DGR sheet provided for validation had a typo (for the date 26-June-2021) which led to showing the energy generation value for the month June-2021 (12-June-2021 to 30-June-2021) to be 11,576.6 MWh. An updated DGR sheet is provided for the entire monitoring period with the generation values and correcting the typo. As the calibration certificates are large in size to attach to the word document, an additional clean copy of the MR is submitted in PDF with the calibration certificates attached to Appendix IV.	
Documentation provided by project participant	
230707_elementsix_Togo_Monitoring_v0.2.xlsx 230705_DGR Sheet_TOGO	
VVB assessment	Date: 21-July-2023
2. PP has now submitted DGR sheet for the current monitoring period. The same was checked and noted that total electricity generation for the current MP is 123,876.40 MWh (column W 16 to W583) which is correctly considered for the ER calculation. VVB further noted that there is a calibration delay from 12-October-2021 to 22-November-2021. Thus, in line with applicable guidance PP has applied the maximum permissible error for the concerned period which is appropriate. Thus, the actual energy generation is 123,855.67 MWh and calculated ER value is 70,969 tCO₂. The submitted VCS MR version 4 dated 17-July-2023 in section 1.1, 1.11 mentions these values as 123,876.4 MWh and 70,969 tCO₂. Thus, a correction is required in the value for energy generation in the VCS MR to avoid mismatch in the data between ER sheet and MR. VVB noted that ER mentioned in the said sections are based on 123,855.67 MWh. CL is open.	
Project participant response	Date: 21-July-2023
Sections 1.1 and 1.11 have been updated to reflect the changes requested by the VVB on the generation values. The corrected value (123,855.67 MWh) relevant to the 70,969 tCO ₂ can be seen now in those sections. The mismatch with the ER is now avoided.	
Documentation provided by project participant	
230721_BLITTA_TOGO_Monitoring Report	
VVB assessment	Date: 21-July-2023
Revised MR was checked and found appropriate. CL is closed.	

CAR ID	1	Section no.	4.1	Date: 10-June-2023
Description of CAR				
Submitted MR is prepared in version 4.1. The latest VCS MR template version is 4.2. Corrective action sought.				
Project participant response				Date: 14-June-2023
The MR is updated to the latest version of template.				
Documentation provided by project participant				
230614_BLITTA_TOGO_Monitoring-Report				
VVB assessment				Date: 22-June-2023
The submitted VCS MR version 03 dated 14-June-2023 is prepared in VCS template version 4.1 Corrective action sought. CAR is open.				

Project participant response	Date: 07-July-2023
The submitted VCS MR version 04 is prepared in VCS template version 4.2.	
Documentation provided by project participant	
230712_BLITTA_TOGO_Monitoring Report	
VVB assessment	Date: 21-July-2023
The latest VCS MR version 4 was checked to confirmed that it is prepared using the most recent VCS template i.e. 4.2. Hence this CAR is closed.	

CAR ID	2	Section no.	4.1	Date: 10-June-2023
Description of CAR				
1. Section 4.2 , table 1 mentions serial nos. of the energy meters which were not in line as confirmed during the site visit. Corrective action sought. 2. In the ER sheet submitted, PP to provide electricity generation data as per the invoices also to show comparison between the values as per energy meter readings and invoice values.				
Project participant response				Date: 14-June-2023
1. In the calibration certificates of the meters, it can be observed that there are two serial numbers provided (Manufacturer Serial Number and Owner Serial Number) for the same energy meter. The one observed during the site visit is the Owner Serial Number while the one provided in the MR is Manufacturer Serial Number. 2. The electricity generation data used for the ER calculations are in accordance with the energy meter readings and invoices.				
Documentation provided by project participant				
Revised MR and calibration certificates				
VVB assessment				Date: 22-June-2023
1. PP response is accepted. The same is confirmed based on the submitted calibration certificates by the assessment team. Hence this part of CAR is closed. 2. PP response is unclear. In the ER sheet submitted, PP to provide electricity generation data as per the invoices also to show comparison between the values as per energy meter readings and invoice values. This part of CAR is open. CAR is open.				
Project participant response				Date: 07-July-2023
2. ER sheet is updated to show the values of both; the daily electricity generation as per the Daily Generation Report (DGR) sheet and the monthly invoices.				
Documentation provided by project participant				
230707_elementsix_Togo_Monitoring_v0.2.xlsx				
VVB assessment				Date: 21-July-2023
The submitted ER sheet was checked and the values were confirmed against the monthly invoices submitted and found appropriate. CAR is closed.				

CAR ID	3	Section no.	4.1	Date: 30-June-2023
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Description of CAR	
1. In line with VCS Monitoring report version 4.2 template guidelines; a. In section 1.1 Provide a summary description of the implementation status of the project for no more than one page. The current information exceeds the same. b. In section 1.5, only indicate the project start date, specifying the day, month and year c. In section 1.7, PP to provide information which is relevant to the project location only. Moreover, VVB checked the provided geographical c-ordinate in software – google map which shows the project location whereas the image provided in the MR seems to be old. d. In section 1.11, table 1, row 1, MR mentions SDG target as 3.8 whereas registered VCS PD mentions the same as 3.1. In row 3, MR mentions 7.b whereas registered VCS PD Mentions the same as 7.1. In row 5, MR mentions 13.0 whereas registered VCS PD Mentions the same as 13.1. e. In section 2.2, PP to provide details on “How due account of all and any input received during ongoing communication has been taken. Include details on any updates to the project design or justify why updates are not appropriate.” f. In section 4.2, the unit of measurement of the parameter is inconsistent with the registered VCS PD. g. In section 4.3, PP to provide “the procedures used for handling any internal auditing performed and any non-conformities identified” 2. PP has submitted monitoring report however, in submitted MR version 3, from page no. 13 onwards, the header mentions that the submitted document is Project Description.	
Project participant response	Date: 07-July-2023
1. a. The summary description has been contained to one page as per the template requirements. b. Section 1.5 has been updated to only indicate the project start date, specifying the day, month and year. c. Section 1.7 has been updated to only provide relevant information regarding the project location. An updated image has been provided from the software Google Maps. d. The numbers (3.1, 4.1, 7.1, 8.1 and 13.1) mentioned in the VCS PD are serial numbers for each subcategory of the monitoring parameters (as shown in last column of table created in section 1.17 of VCS PD) as no template was provided for the SDG Contributions in the VCS PD and since no SDG target numbers were asked by the VCS PD template. The numbers mentioned in the MR (3.8, 4.1, 7.b, 8.1, and 13.0) are the SDG Target numbers as required by the Monitoring Template 4.2 (as shown in column 2 of table provided in section 1.11 of the MR). e. The details on “How due account of all and any input received during ongoing communication has been taken. Include details on any updates to the project design or justify why updates are not appropriate” is provided in section 2.2 of the submitted VCS MR. f. The unit of measurement has been updated in accordance with the registered VCS PD. g. ‘The procedures used for handling any internal auditing performed and any non-conformities identified’ is added to section 4.3. 2. The header has been changed to Monitoring Report: VCS Version 4.2 throughout.	
Documentation provided by project participant	
230712_BLITTA_TOGO_Monitoring Report	
VVB assessment	Date: 21-July-2023

1. Revised MR version 4 was checked to confirm the following; a. Section 1.1 is in line with template requirements and hence accepted. b. Section 1.5 is now revised in line with template requirements and hence accepted. c. Section 1.7 is now revised in line with template requirements and hence accepted. d. PP response is accepted. e. Section 2.2 is now revised in line with template requirements and hence accepted. f. Section 4.2 is now revised in line with template requirements and hence accepted. g. Section 4.3 is now revised to add 'deviation control mechanism' which acts as an internal auditing procedure and hence accepted. 2. Revised MR version 4 was checked and found ok. CAR is closed.
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CAR ID	4	Section no.	4.1	Date: 30-June-2023
Description of CAR				
PP has submitted calibration certificate and calibration schedule is presented in appendix IV of the submitted MR version 3. The project is commissioned in June, 2021. The calibration frequency of the energy meters is annual as per section 4.2. As per appendix IV, all the energy meters were 1 st calibrated on 12-October-2020 and the next calibration was conducted on 22-November-2021. Thus, VVB observed that there is a delay of 1 month as per frequency mentioned in the MR and error factor is to be applied to electricity generation for the delayed period in line with para 366 (a) of CDM validation and verification standard for project activities, version 3.0.				
Project participant response				Date: 12-July-2023
As the first calibration of the meters took place on 12-October-2020 and the second one on 22-November-2021, there is a delay of 42 days (including both dates). According to the 'CDM validation and verification standard for project activities', 366(a), if the meters' calibration is delayed, then the manufacturer's maximum permissible error must be applied to the metered values for that period, provided the error observed during calibration is below the maximum permissible error. The calibration certificates provided in the MR show that the error observed during calibration was below the maximum permissible error which is $\pm 0.2\%$ for 0.2S class meters. The sheet 'Corrected Values' in the '230707_elementsix_Togo_Monitoring_v0.2' shows this error being applied to the delayed 42 days. The difference is then subtracted from the total actual generated value. Therefore, the total corrected emission reductions considered for this monitoring period is 70,969 tCO ₂ e, compared to the original total emission reductions of 70,980 tCO ₂ e.				
Documentation provided by project participant				
230707_elementsix_Togo_Monitoring_v0.2.xlsx				
VVB assessment				Date: 21-July-2023
PP response is accepted and detailed assessment is already considered as a part of CL 01 and hence not repeated here. CAR is closed.				

CAR ID	5	Section no.	ER sheet	Date: 30-June-2023
Description of CAR				

1. PP has now provided “Energy - Generation Monitoring Reports (MWh)” in column C of the submitted ER sheet. Same were cross checked against date presented in column I of the worksheet – Yearly data of submitted excel “DGR sheet – TOGO”. VVB noted that there is a difference of values between the two sources: E.g. for the month of June, 21, ER sheet provides value as 3,811.73 MWh whereas the ‘yearly data’ mentions this value as 13146.5 MWh. VVB observed that the values are mismatching for all the months presented. 2. The current monitoring period is up to 31-December-2022 as per the ER sheet and MR version 3. The submitted DGR data as per the file - “DGR sheet – TOGO” is up to July 22.	
Project participant response	Date: 07-July-2023
1. The DGR sheet (DGR sheet_TOGO) that the VVB is referring to was provided in July 2022 for validation purposes only. It was not submitted as a part of the verification and hence was not the updated version. The previous DGR sheet provided (DGR sheet_TOGO) for validation had a typo which led to showing the energy generation value for the month June-2021 (12-June-2021 to 30-June-2021) to be 11,576.6 MWh. The yearly data sheet in the previous DGR sheet also considered electricity generation from 05-June-2021 to 30-June-2021 for the month June which led to showing the value to be 13,146.5. An updated DGR sheet is provided with the generation values and correcting the typo, and also includes electricity generation from only 12-June-2021 to 30-June-2021 in the ‘yearly data’ sheet. The more conservative values of the electricity generation as provided in the DGR sheet are used for emission reduction calculations as advised by the VVB. 2. The DGR sheet that VVB is referring to was provided only for validation purposes and not submitted as a part of verification. Since it was submitted as part of validation, it was only updated until July 2022. An updated DGR is submitted that provides values up to 31-December-2022.	
Documentation provided by project participant	
230707_elementsix_Togo_Monitoring_v0.2.xlsx 230705_DGR Sheet_TOGO	
VVB assessment	Date: 21-July-2023
PP has now submitted DGR sheet which is required as a part of verification which was checked and confirmed. The monthly energy generation values were cross checked with invoices and were found okay. CAR is closed .	

APPENDIX 3: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Verification team member

No.	Role	Type of resource	Last name	First name	Affiliation (e.g. name of central or other office of DOE or outsourced entity)	Involvement in			
						Desk review	On-site inspection	Interviews	Verification findings
1.	Lead Auditor/Technical Expert	OR	Pundlik	Deepak	TQC-Outsourced entity	Yes	Yes	Yes	Yes

Technical reviewer and approver of the verification and certification report

No.	Role	Type of resource	Last name	First name	Affiliation (e.g. name of central or other office of DOE or outsourced entity)
1.	Technical reviewer (TR)	EI	Meesa	Srikanth	Applus+ Certification
2.	Approver	IR	Calle de Miguel	Agustín	Applus+ Certification

Short CVs of the Team:

- Mr. Deepak Pundlik** has experience in climate change, waste management and environmental management. After completing Masters in Environment Sciences from Pune university, He has worked in waste management field. As a GHG consultant, he handled more than 50 projects under renewable energy, waste management sectors during his stint with companies - MITCON and Thermax Limited. As a GHG auditor, he has validated/verified projects under CDM/VCS/GS and GCC mechanisms from renewable energy, energy demand, waste management sectors.
- Mr. Srikanth Meesa** is a climate change professional with more than 17 years of experience in the fields of climate change, GHG auditing, carbon footprint assessments, water and energy audits, scope-3 emissions, net zero strategies, sustainability & ESG sectors and water

and wastewater treatment sectors. He has completed his masters in Environmental Engineering& Management from I.I.T Delhi and Bachelors in Civil Engineering from Osmania University. Furthermore, he has extensive experience i.e., more than 14 years on the auditing of CDM, VERRA, GS, JCM projects of various sectoral scopes across the world. He has worked with reputed certification bodies such as TUV NORD, DNVGL, TUV SUD and LR. He also has consulting experience on sustainability reports& road-maps, stakeholder consultation and LCA studies while working at Thinkstep (Now called Sphera). Currently, he is associated with Global Green Solutionz and empanelled with Applus+ Certification to carry out GHG audits in the aforementioned schemes. Mr. Srikanth Meesa is based at Hyderabad, India. Mr. Srikanth Meesa participates as part of the Audit Team as Technical Reviewer for the assessment.

APPENDIX 4: ABBREVIATIONS

Abbreviations	Full texts
BM	Build Margin
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification Request
CM	Combined Margin
CO ₂	Carbon Dioxide
CO ₂ e	Carbon Dioxide equivalent
DNA	Designated National Authority
DR	Document Review
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reductions
FAR	Forward Action Request
GHG	Greenhouse gas(es)
GWP	Global Warming Potential
PP	Project Participant
VCU	Voluntary Carbon Unit
VCS	Voluntary Carbon Standard
VVB	Validation and Verification Body

APPENDIX 5: CALIBRATION DETAILS

Sr. No.	Serial numbers of Energy meters	Accuracy class	Calibration schedule
1	56198213	0.2s	12-October-2020 22-November-2021 25-October-2022
2	56198214	0.2s	12-October-2020 22-November-2021 25-October-2022
3	56198215	0.2s	12-October-2020 22-November-2021 25-October-2022
4	56198216	0.2s	12-October-2020 22-November-2021 25-October-2022
5	56198203	0.2s	12-October-2020 22-November-2021 25-October-2022
6	56198217	0.2s	12-October-2020 22-November-2021 25-October-2022
7	56198221	0.2s	12-October-2020 22-November-2021 25-October-2022
8	56198222	0.2s	12-October-2020 22-November-2021 25-October-2022